

Correction note

Prepended 21 September 2026 to the record as originally rendered, which follows unchanged.

Correction note — 21 September 2026 (D-42). This record's cold-start baseline, 442 / 259 / 183 / 0, was inflated by a classifier defect: API batches cut off at the token limit were cached as if the model had declined to classify them. The corrected figures for the same 442 people are **314 classified, 128 needing review (29.0%), 0 not-an-occupation**. The gate's verdicts stand: it tested that every surface derived the figure from one source and agreed through a correction sequence, and they did. What was wrong was the input to that derivation, which no consistency check can detect. The record below is left as written. See `docs/d42_failure_record.md` for the cause, the evidence and the fix.

Gate D — counter integrity

Day 18. Closed 14 of 14. One continuous browser session, no restart, no cache clear, no crash. No fixes applied inside the gate.

Definition of done

The gate plan of 2 September defined Gate D in a single line: *the needs-review figure verified live-derived at every surface it appears*. The surfaces were never enumerated, so the enumeration below was proposed, agreed and then produced by source-level sweep before the gate opened.

Full DoD as run: every surface rendering a population count is enumerated by source-level sweep; each is verified live-derived from a single source with corrections applied; all agree across a correction sequence; the three display states close on the population at every step; a written verdict per surface; no fixes inside the gate.

The distinction the gate turns on is **agreement versus derivation**. Two surfaces computing the same number by different routes agree until one route changes. That is the D-15 shape, and it is not visible in a screenshot — which is why steps 1-4 are source-level and only step 14 tests the claim at runtime.

Baseline

442 people · 259 classified · 183 needs review · 0 not-an-occupation.

Verdicts

#	Check	Result
1	<code>sweep_count_surfaces --ui-only --strict</code> : 26 count-bearing strings across 13 files, 0 unexempt	PASS
2	<code>check_single_loader</code> : 7/7	PASS
3	<code>pytest</code> : 267	PASS
4	Five count sources: 4x <code>coverage_counts</code> , 1x <code>review_counts</code> for the baseline M	PASS
5	Cold start, five surfaces, four suppression branches fire	PASS
6	259 + 183 + 0 = 442; 240 + 19 = 259	PASS
7	One assignment: 260 / 182, M held at 183	PASS
8-9	Four assignments + one not-an-occupation: 263 / 178 / 1	PASS
10	Two skips: queue view loses two, no count moves	PASS
11	Query narrows, clears, group filter — sidebar moves alone	PASS
12	Download and restore: identical counts, M = 183, report reads 5 of 5	PASS
13	Second restore of the same file: idempotent	PASS
14	Sixth correction: 264 / 177 / 1, all surfaces move in ONE rerun	PASS

Surfaces enumerated

Seven. Six observed live.

Surface	Site	Count source	Verdict
Header	<code>app.py:167/169</code>	<code>coverage_counts</code> (N), <code>review_counts</code> (M)	live-derived
Sidebar counter	<code>search.py:71/72</code>	<code>len(matches) / len(df)</code> — a filter counter, not a population count	correct by scope
Summaries caption	<code>summaries.py:282/285/288/298/301</code>	<code>coverage_counts</code>	live-derived
Chat caption	<code>assistant.py:503/506/508/510</code>	<code>coverage_counts</code>	live-derived
Review queue	<code>people_list.py:158/159</code>	<code>coverage_counts</code>	live-derived
Pager	<code>people_list.py:270</code>	page arithmetic	exempt, confirmed live
Restore report	<code>session_io.py:159/163</code>	restore result	exempt; L159 observed, L163 not

N — the needs-review figure — now has exactly one derivation across the four population surfaces. M is the session's fixed starting backlog and legitimately takes the second route: it counts `is_uncertain` on the pre-correction base and cannot be recovered from a frame that already has corrections applied.

Exemptions

Eight N of M constructions carry inline `# count-exempt:` markers. The sweep detects the *construction*, not the guard, so a correctly guarded site still matches; the reason field states which of three cases applies.

Site	Reason	Class
<code>app.py:169</code>	guarded, n != m	guarded
<code>people_list.py:159</code>	guarded at L154 (D-32)	guarded
<code>search.py:72</code>	guarded by <code>filtering</code> ; the filtered-and-matching-all case is intended	guarded
<code>summaries.py:288</code>	guarded at L282/285	guarded
<code>assistant.py:506</code>	guarded — equal case handled (D-34)	guarded
<code>people_list.py:270</code>	<code>_render_pager</code> returns early when <code>total_pages <= 1</code>	unreachable

session_io.py:159	equality is the reassurance — "5 of 5 corrections" is the point	intended
session_io.py:163	as above, for notes	intended

Two were confirmed by observation rather than argument: the pager rendered 11-20 of 442 on a non-first page, so its equal case genuinely cannot occur; and the restore report rendered 5 of 5, which is the reason working as stated. session_io.py:163 never rendered — this session carried no notes — so its reason remains reasoned but unobserved.

Defects closed before the gate

		Commit
D-32	Review queue rendered "183 of 183" at cold start	38d3a63
D-34	Chat caption would render "all 442 classified connections of 442"	4769cfe
—	Sidebar rendered "442 of 442 shown" at rest	2cc7542
D-33	Header read the node table a second time off disk	cb370e9 , 43e51b2
D-35	Review queue counted display_state inline — a third implementation	—
D-36	Sidebar caption extracted to a pure filter_caption()	—
D-37	Checker exemptions moved to the site, checked per line	—
D-38	Double-encoded em-dashes in check_single_loader output	in 43e51b2

D-33 is the one that mattered. app.py called get_review_counts() with no argument, which read network_nodes.csv off disk a second time through a separate cached function, ignoring the frame load_display_table() had assembled one line earlier. Every count agreed, so nothing on screen said so. At P2.9 the upload flow hands the loader its own pipeline output — so the moment a user uploaded their network, the four display_state surfaces would have reported their counts while the header reported whatever sat in data/graph/network_nodes.csv on the server. After P2.9a that is the synthetic demo set. Not a crash: a confidently wrong number in the largest type on the page, beside four correct ones.

tests/test_review_count_equivalence.py was written against unchanged code and verified green at ad7ef5a with both routes live, then green after the collapse. D-33 is therefore a proven refactor, not an assumed one. The structural proof is that check_single_loader's exemption for state.py — "owns NODES_CSV as the base read" — could be deleted. That exemption was the defect, and it hid it for exactly as long as it stood.

D-34 was unreachable by observation. Its trigger requires every person classified, which no screenshot of the 442-row table can produce. It was found by source-level sweep and closed by unit test on a synthetic frame.

Instrumentation built for this gate

- tools/sweep_count_surfaces.py — AST enumeration of count-bearing strings, inline # count-exempt: markers read via tokenize, --ui-only and --strict. AST rather than grep for the reason check_single_loader learned: a docstring mention produces a false positive.
- check_single_loader.py criterion 1 — now per-line with site-local # loader-exempt: markers, replacing a file-level exemption dict.
- tests/test_review_count_equivalence.py — seven correction scenarios, four properties, pinning the two derivations against each other.

Known limits

- strict cannot catch a stale reason. It fails on new unmarked constructions only. Delete a guard, leave the marker, and the check stays green. That is check_single_loader's state.py exemption one level up. The mitigation is that every reason prints in the report and is read at the gate — a human step, not a guarantee. This fired once during D-35: an edit moved a guard and left the marker's line number stale, and only reading the reason caught it.
- Two exemption mechanisms, two matching rules. count-exempt matches a span of [lineno-1, end_lineno]; loader-exempt requires an exact line match. Logged as D-39.
- session_io.py:163 unobserved, as above.
- Sidebar counter is scoped differently from the other four. It counts filter matches, not display states, and does not exclude needs-review people. Correct, but it is not a population count and should not be read as one.

Open at close of gate

D-11 (Phase 4), D-21, D-22, D-29 (docs), D-37 markers' wording, D-39 (reconcile the two marker rules), D-40 (both rebuild-diff tools fail on a clean checkout because .bak430 is gitignored; state it in their docstrings).

Next

Phase 2 checkpoint: P2.9a (synthetic 442-row demo set, self-node identity), D-31 (build/runtime manifest split, verified in a clean venv), P2.9 (deploy, README, demo, zero-crash).